Infosys - Coding 19

HackerRank

In satellite signal processing, a stream of N data packets is received. Each packet i is classified into a category A[i] (ranging from 0 to C−1) and carries a specific signal weight W[i] representing its data density.

A monitoring window [l, r] (the contiguous range of packets from index l to r) is considered frequency-balanced if there is at least one category that appears in strictly more than half of the packets in that window. In other words, some category is a strict majority in the window.

Find the maximum total weight (the sum of W[i] for all packets in the window) over all possible frequency-balanced windows.

Input Format

First line: integer N, the number of data packets.

Second line: integer C, the number of possible categories.

Next N lines: integer A[i] (for 0 ≤ i < N).

Next N lines: integer W[i] (for 0 ≤ i < N).

Constraints

1 <= N <= 10^5

1 <= C <= 10^5

-10^9 <= A[i] <= 10^9

-10^9 <= W[i] <= 10^9

Output Format

The output contains an integer value

Sample Input 0

```
3
3
1
2
1
10
10
10
```

Sample Output 0

```
30
```

Explanation 0

Selecting the entire window of three packets allows category 1 to appear twice (2 > 3/2) to form a strict majority, yielding a total accumulated weight of 30.

Sample Input 1

```
3
4
1
2
3
100
100
100
```

Sample Output 1

```
100
```

Explanation 1

With completely unique categories 1, 2, and 3, no category can achieve a strict majority in any window larger than a single packet, making the optimal choice any individual packet with its maximum weight of 100.